PROJECT DOCUMENT

# StudyMate AI — Your Personal AI Study Companion

StudyMate AI is an intelligent web-based learning assistant that helps students turn notes and documents into focused, exam-ready study materials. It reduces information overload by using AI to summarize content, explain difficult concepts, generate likely exam questions, build flashcards, highlight key points, and visualize ideas as mind maps.

## Project Overview

StudyMate AI is designed to improve the way students prepare for tests and examinations. Instead of reading long notes repeatedly, learners can upload a document or paste text and instantly receive structured outputs that support faster understanding and more effective revision. The platform combines artificial intelligence with a student-friendly interface to make learning clearer, quicker, and more interactive.

### Problem the Project Solves

Many students struggle with bulky notes, complex textbook explanations, limited revision time, and uncertainty about what is most important for examinations. This often leads to poor concentration, weak retention, and inefficient study habits.

### Target Users

The platform is built for secondary school students, university students, and exam candidates preparing for assessments such as WAEC, JAMB, and similar academic tests.

## Project Objectives

- Help students understand difficult topics in simple and familiar language.
- Reduce long reading time by turning large notes into concise revision materials.
- Improve exam readiness through question practice and focused review content.

- Support active recall with flashcards and key revision points.
- Make study sessions more engaging, structured, and productive.

## Core Features

- **Summarize:** Produces a short, clear summary from long notes or documents.
- **Explain:** Breaks difficult concepts into simple teacher-style explanations.
- **Generate Questions:** Creates 20 multiple-choice questions and 10 theory questions for revision.
- **Build Flashcards:** Generates 12 flashcards to support recall and spaced repetition.
- **Key Points:** Extracts the main areas of concentration and the most important facts to remember.
- **Concept Tree:** Converts content into a visual mind map using Mermaid diagrams.
- **History Tracking:** Saves previous study results locally for future review.
- **PDF Export:** Allows users to export generated study output as a PDF document.
- **Voice Support:** Includes voice input and text-to-speech for more accessible study sessions.
- **Document Upload:** Accepts both typed notes and uploaded files such as PDFs and images.

### How the System Works

- The student types notes or uploads a document.
- The user selects a preferred AI study action.
- The system sends the content to the AI engine for processing.
- The platform returns a structured learning output for study and revision.
- The result can be copied, listened to, saved in history, or exported as PDF.

### Expected Benefits

- Better concentration during revision.
- Improved understanding of difficult topics.
- Faster preparation for exams and tests.
- More engaging and personalized study sessions.
- Better knowledge retention through repeated active recall.

## Technology Stack

- **Frontend:** Next.js 16, React 19, CSS
- **AI Integration:** Google Generative AI (Gemini API)
- **Visualization:** Mermaid for concept trees and mind maps
- **Document Export:** jsPDF and html2canvas
- **Storage:** Browser local storage for history and progress tracking

## Security and Privacy Considerations

StudyMate AI includes dedicated privacy, security, and service status pages. The platform is structured to process user inputs for study assistance while retaining history and preferences primarily in the user's browser local storage. This approach supports a more privacy-conscious learning experience and gives users visibility into application trust and service availability.

## Possible Future Improvements

- Personalized study plans and revision schedules.
- Topic mastery scoring and adaptive recommendations.
- Collaboration features for group study.
- Offline access for previously generated materials.
- Support for more file types and multilingual study assistance.

## Conclusion

StudyMate AI is a practical educational technology solution that helps students study smarter, understand faster, and prepare better for examinations. By converting raw learning material into clear summaries, simple explanations, quizzes, flashcards, key points, and visual concept maps, the platform provides a complete AI-powered revision experience in one place.

---

Prepared for the StudyMate AI project based on the implemented application features and user experience available in the current codebase.